

Neural dynamics of pain modulation by emotional valence

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Definitions of human pain acknowledge at least two dimensions of pain, affective and sensory, described as separable and thus potentially differentially modifiable. Using electroencephalography, we investigated perceptual and neural changes of emotional pain modulation in healthy individuals. Painful electrical stimuli were applied after presentation of priming emotional pictures (negative, neutral, positive) and followed by pain intensity and unpleasantness ratings. We found that perceptual and neural event-related potential responses to painful stimulation were significantly modulated by emotional valence. Specifically, pain unpleasantness but not pain intensity ratings were increased when pain was preceded by negative compared to neutral or positive pictures. Amplitudes of N2 were higher when pain was preceded by neutral compared to negative and positive pictures, and P2 amplitudes were higher for negative compared to neutral and positive pictures. In addition, a hierarchical regression analysis revealed that P2 alone and not N2, predicted pain perception. Finally, source analysis showed the anterior cingulate cortex and the thalamus as main spatial clusters accounting for the neural changes in pain processing. These findings provide evidence for a separation of the sensory and affective dimensions of pain and open new perspectives for mechanisms of pain modulation.

Key words: emotion; pain; electrical stimuli; somatosensory evoked potentials; priming.

Introduction

Pain signals a threat and motivates behavior to protect the organism from injury. Following the International Association for the Study of Pain terminology, pain is an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage (Raja et al. 2020). Pain can therefore be defined by a sensory-discriminative (intensity), an affective-motivational (unpleasantness) and a cognitive-evaluative dimension (Melzack and Casey 1968; Fernandez and Turk 1992).

Pain perception can be modulated by various emotional and cognitive factors (Keefe et al. 2001; Meagher et al. 2001; Rhudy and Meagher 2001; Villemure et al. 2003; Rhudy et al. 2005). For instance, emotions have been repeatedly shown to alter pain perception (de Wied and Verbaten 2001; Meagher et al. 2001; Kenntner-Mabiala and Pauli 2005; Kenntner-Mabiala et al. 2008).

Emotion is usually conceptualized along two main dimensions: valence (pleasantness/unpleasantness) and arousal (calm/excited) (Russell 1980). The mechanisms by which emotions interact with the dimensions of pain remain largely unclear (Bushnell et al. 2013). Lang's motivational priming hypothesis states that when

a certain emotion is primed or activated, the responses of the system subserving the emotion in question will be facilitated, whereas the responses of the opposed system will be inhibited (Lang 1995). In line with this theory, several studies reported consistent reduction of perceived pain when simultaneously administered with the presentation of positive stimuli (de Wied and Verbaten 2001; Wunsch et al. 2003), and increased perceived pain during presentation of negative stimuli. Neutral stimuli usually did not show modulatory effects on pain perception (Rhudy et al. 2010; Kamping et al. 2013).

Important distinctions have also been made regarding these emotional modulatory effects for different dimensions of pain, namely pain intensity and pain unpleasantness, (Price et al. 1987; Zelman et al. 1991; Fernandez and Turk 1992; Auvray et al. 2010). For example, attention was shown to play an important role for the modulation of pain intensity but not for pain unpleasantness (Kenntner-Mabiala et al. 2008). This separable dimensionality of pain is also apparent in some chronic pain states. For example, in fibromyalgia patients, pain intensity and pain unpleasantness were differentially modulated by emotional valence (Kamping et al. 2013). Moreover, cancer patients reported more

unpleasantness associated with their pain than did women in labor while the sensory dimension of pain was rated similarly in both pain groups (Price et al. 1987).

Despite such evidence, the dissociation between pain intensity and unpleasantness is still debated (Chapman et al. 2001; Horn et al. 2012). Some authors argue that only the affective, and not the sensory dimension of pain can be modulated using emotional or cognitive manipulation (for review [Talbot et al. 2019]).

There are various experimental factors that may explain different outcomes between studies. For example, various assessment of pain ratings. Some studies collected only one pain rating and showed a modulatory effect on pain intensity (the sole scale employed for assessing pain perception), e.g. (Kenntner-Mabiala and Pauli 2005; Godinho et al. 2006). Notably, such outcome could occur primarily as a result of collecting a single pain rating. In practice, these ratings may also include the emotional dimension, creating an illusion of modulation of pain intensity that may in reality be influenced by pain unpleasantness (Chapman et al. 2001).

In addition, the modality of the prime being used (sound, words, pictures) is also of importance. Priming effect of pictures has been shown to be stronger than the priming effect of sounds or of words, especially for short interstimuli intervals (~500 ms) (Bajo 1988; Chen and Spence 2010; De Lucia et al. 2010; Kim et al. 2014), possibly because pictures have an automatic semantic representation whereas auditory stimuli or words might require more time to access associated semantic representations. For these reasons, pictures are believed to elicit a strong unconscious priming effect, which motivated the design used in the present study (McBride and Doshier 2002; Spruyt et al. 2007).

Moreover, the duration of the presentation of the emotional stimuli might play an important role. Most studies used relative long emotional picture durations (>6 s), other studies used shorter ones or subliminal (33 ms) masked stimuli (Kenntner-Mabiala et al. 2008; Richter et al. 2014; Pelaez et al. 2016). Prolonged prime exposure might result in increased recruitment of consciously controlled processes (Meier et al. 2003). It is indeed acknowledged that short prime durations might involve an automatic component of emotional processing, while longer picture presentation might involve explicit cognitive processes such as attention or expectation (Jacoby 1991). Specifically, semantic representations were shown to take place between 200 and 300 ms, followed by attentional processes between 300 and 400 ms after stimulus onset (Hauk et al. 2008; Richter et al. 2014).

Interestingly, shorter primes that are too brief to be consciously perceived resulted typically in stronger effects compared to long-duration primes (Stapel and Koomen 2005; Rieth and Huber 2010; Brunet 2023). Furthermore, research indicated enhanced modulatory effect for short compared with long interstimulus intervals between prime and painful stimuli (<500 ms) (Richter et al. 2014).

Enhanced modulatory effect on pain have also been reported for clearly painful electrical stimulation but not for nonpainful somatosensory stimuli (Godinho et al. 2006; Montoya and Sitges 2006).

In the present study, in order to recruit automatic semantic processing (where modulation is assumed to be without awareness of the participant), we presented short prime pictures (200 ms) with various emotional contents (negative, neutral, positive), followed by application of electric painful stimuli with an interstimulus interval of 200 ms. After each painful stimulus, we assessed both perceived pain intensity and unpleasantness. We also recorded

neural activity using electroencephalography during the entire experiment.

In event-related potential (ERP) studies, pain has been reported to elicit a biphasic negative–positive wave with latencies within 100–300 ms after pain onset (for a review, [Baumgartner et al. 2012]). The first commonly reported pain-related negative potential called N1 or N2, depending on the pain stimulation method used, has been described to mainly reflect sensory aspects of perceived pain (Iannetti et al. 2005). The role of N1/N2 is however still debated, and has been associated with nonnociceptive processing, and to be rather involved in the saliency of the eliciting stimulus, independent of its modality (Iannetti et al. 2008; Mouraux and Iannetti 2009). The cortical sources of the N1/N2 have been localized in operculo-insular areas and primary somatosensory cortex (Iannetti et al. 2005). The source of the latter positive (P1 or P2) has been indicated to be generated in the anterior cingulate cortex (ACC; Lenz et al. 1998; Kakigi et al. 2000; Garcia-Larrea et al. 2003) or thalamus (Kakigi et al. 1989; Kakigi et al. 2000). The identification of cortical sources varies, however, across studies, possibly because of the stimulation methods used, for example, electric vs. laser stimuli (Kanda et al. 2000). Studies using laser stimuli reported difficulties to show reliable S1 responses in contrast to studies using noxious electrical stimuli, possibly due to the increased S1 response to nonnociceptive afferents (skin, peripheral nerves) for electrical stimuli (Kitamura et al. 1995; Garcia-Larrea et al. 2003).

The identification of cortical sources when pain is modulated by emotional stimuli has been little investigated. Some activations clusters have however been consistently reported; such as the bilateral insula and temporal poles, parietal and prefrontal cortices (PFC; Godinho et al. 2006; Stancak and Fallon 2013). Neural activation in these areas is also consistent with brain imaging studies on modulation of pain by emotions (Roy et al. 2009; Kamping et al. 2016). Additional areas such as the ACC and thalamus have also been involved in the circuitry underlying emotional modulation of pain, but not systematically (Villemure and Bushnell 2009; Stancak and Fallon 2013). Inconsistencies in such findings could result from the relatively small sample size, albeit standard for EEG or brain imaging studies ($n < 16$), which might affect statistical power. In addition, these studies examined cortical sources at different time windows (i.e. pre-, during or postpainful stimuli; e.g. 270–400 ms in Godinho et al. (Godinho et al. 2006); –800 to 700 ms in Stancak et al. (Stancak and Fallon 2013), where different processes might be taking place and might therefore involve the recruitment of different brain areas.

We hypothesized that pain intensity and pain unpleasantness would be differentially modulated by our affective manipulation, such that pain unpleasantness ratings would be reduced when preceded by positive emotions, and enhanced when preceded by negative emotions. We expected pain intensity ratings to reflect mainly the sensory quality of pain, and therefore to remain constant across conditions because the pain stimuli were all applied at the same intensity.

For the same reasons, we did not expect differences in N2 amplitudes across valence conditions. In contrast, the later positive potential P2 was expected to mainly reflect the affective dimension of pain, with higher amplitudes for negative compared to positive and neutral emotions in between.

Furthermore, we expected the source of the ERP signal to be located in regions relevant for pain intensity such as thalamus, somatosensory cortices, and in regions relevant for affective processes such as ACC and PFC (Treede et al. 1999; Price 2000; Apkarian et al. 2005).

Materials and methods

Participants

Twenty-one healthy volunteers (11 females, mean age = 23.53 years, $SD = 2.55$ years) participated in the experiment upon completing a written consent for taking part in the study. They also filled in the Edinburgh Handedness Inventory and a short demographic checklist. All participants were right handed (assessed using the Edinburgh Handedness Inventory [Oldfield 1971]). Inclusion criteria were normal or corrected to normal vision and no history of mental or neurological disorders. The data of two participants were subsequently excluded from the EEG analysis due to less than 75% artifact free epochs. Thus, the final sample consisted of nineteen participants. The study adhered to the Declaration of Helsinki and was approved by the Ethics Committee of the Medical Faculty Mannheim, Heidelberg University, Germany.

Visual stimuli

Prime pictures comprised 40 negative, 40 neutral, and 40 positive pictures from the International Affective Picture System¹ (Lang et al. 2008). Normative ratings were (pleasant: 7.58, neutral: 5.19, unpleasant: 2.16) for picture valence, and (pleasant: 4.85, neutral: 4.10; unpleasant: 5.77 for arousal, similar to other IAPS studies on evoked-related potentials (ERPs) from the Lang, P. group e.g. (Cuthbert et al. 2000; Keil et al. 2002).

Painful stimuli

electric stimuli were applied at the left forearm by a bar electrode with a constant current stimulator (Digitimer DS7A, Welwyn Garden City, United Kingdom). Electrode position was chosen to minimize muscle activity. Intensity of stimulation was determined for each participant using an individual threshold. Ascending series of electric stimuli were administered starting with a mild stimulus until each participant indicated it as "painful" (pain threshold) and then further until the pain became unbearable (pain tolerance). The entire threshold estimation procedure was repeated three times in order to obtain a reliable measure and only the values of the last two trials were averaged. The chosen stimulus intensity was defined as 80% between pain threshold and pain tolerance such that the sensation was painful but tolerable. Intensity and unpleasantness of the chosen stimulus were then rated on two consecutive numerous analogue scales (NAS). The first scale assessed the perceived intensity of pain (i.e. *How strong was the stimulus?*) from 0 to 10 (0 = no pain and 10 = strongest pain imaginable). The second rated subjective unpleasantness of pain (i.e. *How unpleasant was the stimulus?*) from 0 to 10 (0 = not at all unpleasant and 10 = most unpleasant).

To make sure the calculated intensity was robust and elicited reliable somatosensory sensations, before the experiment started, we applied 10 electrical stimuli at the calculated intensity, and we asked participants to rate how painful the stimuli were using the pain intensity NAS. When the pain ratings did not reach 7 or 8 out of 10 on the NAS scale, stimulus intensities were increased until the desired value was achieved (Lyu et al. 2021).

Experimental procedure

After arrival at the laboratory, participants were seated in a comfortable chair ~50 cm from a computer screen. The stimulation electrodes were then attached to the participant's left forearm and the pain intensity was determined as described above. Then participants went through a practice session to familiarize themselves with the experimental task. During the practice session, participants were trained to minimize eye blinks and movements while attending the display. During the entire experimental session, participants were video monitored in order to ensure that they were looking attentively at the screen.

Each trial began with a fixation cross with a variable duration randomly set between 1,200 and 2,400 ms in order to reduce predictability of the incoming prime picture (Fig. 1). Following the fixation cross, the prime picture appeared for 200 ms, followed by another fixation cross for 1,200 ms. Then, 200 ms after onset of the second fixation cross, the painful electrical stimulus was delivered, followed by two consecutive NASs, prompting participants to evaluate the perceived painful stimulus.

Participants were informed that some pictures would be displayed shortly and be followed by painful electrical stimuli. They were also told that they would have to rate the intensity and the unpleasantness of the painful stimuli. Participants were explicitly made conscious of the conceptual distinction between the sensory and the affective dimension of pain using instructions from the radio example (Price et al. 1983). We carried out as many practice trials as necessary to ensure that the participant understood the task, and that the intensity used was painful but tolerable.

All stimuli were presented on a 17-in. computer monitor over a light gray background. To avoid possible order effects, the order of picture presentation was pseudo-randomized across participants. To counteract sensitization or habituation to the painful stimulation, 3-min breaks were introduced for each participant after every block of 30 trials (Milne et al. 1991).

EEG data acquisition

EEG data were collected using a BrainProducts actiCap (Gilching, Germany) with 64 active Ag-AgCl electrodes. This EEG cap with embedded electrode hubs is based on the extended international 10–10 system. Vertical and horizontal eye movements were recorded with two electro-oculogram electrodes, placed below and at the outer canthi of the right and left eye. FCz was used as reference during recording. EEG signals were digitized at a 1000 Hz sampling rate and amplified via two BrainAmp amplifiers BrainProducts (Gilching, Germany). Electrode impedances were kept below 20 k Ω , as suggested by the manufacturer for this system.

EEG data analysis

EEG data were processed using EEGLAB (Delorme and Makeig 2004), an open source toolbox running under Matlab 8.2 (The Mathworks, Inc., Natick, Massachusetts). The raw data were band-pass filtered (0.1–30 Hz), resampled at 250 Hz, visually cleaned from bad channels, segmented into epochs of 1,100 ms (–200–900 ms around stimulus onset) on the basis of stimulus type (prime picture category) and re-referenced to average reference. A previous independent components analysis (ICA) computed matrix (https://scn.ucsd.edu/wiki/Makoto's_preprocessing_pipeline) was applied to these preprocessed data for artifact removal and extract meaningful components. The matrix was calculated on longer epochs encompassing a time

¹ Picture numbers were: neutral (1390, 1903, 2025, 2032, 2235, 2372, 2487, 2514, 2521, 5900, 6000, 7011, 7013, 7018, 7021, 7033, 7042, 7044, 7057, 7058, 7077, 7081, 7096, 7137, 7140, 7183, 7184, 7188, 7237, 7248, 7249, 7513, 7550, 7560, 7620, 7632, 7820, 7830, 9150, 9468), positive (1410, 2035, 2045, 2050, 2057, 2070, 2150, 2274, 2311, 2340, 2352, 1440, 2360, 2395, 2550, 2660, 4640, 4641, 5220, 5480, 5825, 5830, 1463, 7230, 7260, 7270, 7330, 7470, 8120, 8461, 8496, 8501, 8502, 1510, 8540, 1630, 1710, 1721, 1750, 1999), negative (2301, 2352, 2710, 2800, 2900, 2981, 3016, 3017, 3051, 3059, 3061, 3064, 3168, 3181, 3185, 3220, 3225, 3301, 3550, 6022, 6213, 6243, 6415, 6520, 6560, 6563, 9040, 9043, 9075, 9140, 9181, 9185, 9253, 9265, 9332, 9405, 9571, 9635, 9800, 9902).

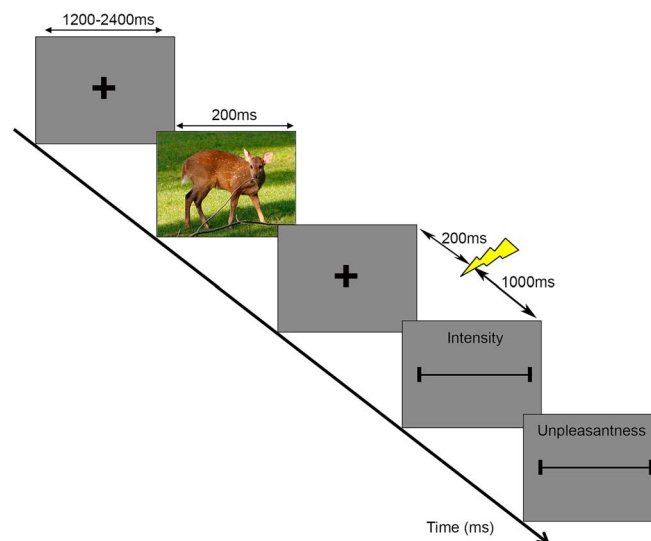


Fig. 1. Schematic representation of the experimental paradigm. Each trial began with a fixation cross with duration between 1,200 and 2,400 ms, followed by a prime picture lasting 200 ms, then another fixation cross was presented for 1,200 ms, during which painful stimuli were applied, starting 200 ms after the prime picture. Then two consecutive NAS scales appeared, on which participants reported the subjective intensity and unpleasantness of the perceived painful stimulus using a keyboard. The prime picture shown is for illustration purposes, it was not used in the study and it is not from the IAPS database.

window of 2,400 ms (1,000 ms before and 1,400 ms postprime picture and band-pass filtered (1 Hz–30 Hz) (Winkler et al. 2015). Missing channels were interpolated using a spherical interpolation algorithm. Semi-automatic artifact correction was performed using ICA (Makeig et al. 1996) and the toolbox SASICA (Chaumon et al. 2015). Components that reflected muscle activity, electrocardiogram, or eye movements, on the basis of their dipole location, spectra and scalp maps were considered as artifacts and excluded from further analysis. Data were baseline-corrected by subtracting the mean of the 200 ms pre-stimulus interval. Data were then processed and visualized using a STUDY structure.

For each individual participant, epochs pertaining to the same experimental condition were averaged in the time domain, time-locked to the onset of the prime picture. This step yielded three average waveforms per scalp site, one waveform for each experimental condition (negative, neutral, and positive prime pictures). A grand-average waveform was then calculated across the single-subject average waveforms.

Electrophysiological source analysis

Independent component source locations were estimated by creating an equivalent current dipole model for each component with the DIPFIT 2.2 (EEGLAB plug-in using Fieldtrip toolbox functions, that estimates dipole location by applying inverse source modeling methods to a standard boundary element head model using the Montreal Neurological Institute (MNI) template (Oostendorp et al. 1989; Rissling et al. 2014). Only components whose scalp maps had <15% residual variance from the best fitting forward model scalp projection were considered for further analysis.

Components were grouped into several clusters with a joint distance measure, on the basis of dipole locations, power, average ERPs, mean Event Related Spectral Perturbation and Inter-Trial phase Coherence measures. These data from each subject were initially decomposed by Principal Component Analysis, and the resulting component distances were clustered with a k-means

algorithm (for further information on this procedure see [Rissling et al. 2014]).

Statistical analysis

Based on visual inspection of the scalp maps (see Fig. 3) and according to previous reports on pain ERPs studies (Iannetti et al. 2005; Kenntner-Mabiala et al. 2008; Ring et al. 2013), N2 and P2 peak amplitudes were extracted in the time windows between 100 and 150 ms and 200 and 300 ms after pain onset. The Cz, C2, C4, and CF2 channels were considered for the N2 deflection; the Cz, C1, C2, CPz, CP1, and CP2 channels were considered for the P2 deflection (Kopp et al. 2020). Statistical analyses were performed in IBM SPSS Statistics for Windows version 23.0 and RStudio software Version 1.3.1093.

Data not normally distributed (as assessed through Shapiro-Wilks test) were logarithmically transformed (base 10). In order to meet normality assumptions required for the following statistical analyses, differential values for each electrode were normalized taking the absolute value followed by log-transformation (base 10). All other assumptions (linearity, multicollinearity, absence of leverage values) were met.

A multivariate analysis of variance (MANOVA) was run to determine the effect of emotional category (negative, neutral, and positive) on electrophysiological (N2 and P2) and perceptual (pain intensity and unpleasantness) mean differences. We chose a multivariate general linear model approach, inclusive of follow up univariate tests, because it is considered as a more robust statistical test for repeated measure designs than using separate ANOVAs (O'Brien and Kaiser 1985).

All of the post-hoc tests were adjusted for multiple comparisons using a false discovery rate correction (Benjamini and Hochberg 1995). Additionally, we examined differences between pain intensity and pain unpleasantness across picture valence using a two-way repeated measures ANOVA.

We also assessed if changes in N2 were correlated with changes in P2 amplitude across picture valence using Pearson's correlations.

For all the perceptual and electrophysiological data analyses, an alpha level of 0.05 was adopted.

We then aimed to assess if differences in the neural correlates of somatosensory stimulation (considering P2, N2, then considering a contribution of both P2 and N2) could linearly predict differences in affective modulation of perceived pain. For this purpose, we carried out a hierarchical regression analysis in order to test which neural predictors (P2, N2, or the complementary contribution of P2 and N2) had an effect on the separated dependent variables. For this analysis, we selected the pain ratings (intensity, unpleasantness) that showed significant differences between the different valence conditions and defined an index of emotional modulation of pain as follows:—We subtracted the unpleasantness pain ratings between negative and neutral, and negative and positive picture valence (resp. $UNP_{neg-neu}$, $UNP_{neg-pos}$).—For ERPs (N2 and P2), since their amplitude was in the log domain, normalization was performed by subtracting ERP amplitudes between valence of interest (e.g. $P2_{neg-neu}$).

Finally we tested the effect of arousal on both ratings and ERP by adding arousal as a covariate in all the above-mentioned analyses.

Results

Multivariate analysis of variance

The MANOVA showed a statistically significant multivariate (overall) effect of the three picture categories (negative, neutral, and positive) on the combined dependent variables (pain ratings and ERP amplitudes), $F(8, 11) = 3.440$, $P = 0.031$; Wilks' $\Lambda = 0.286$; partial $\eta^2 = 0.714$.

Perceptual results

Follow-up univariate ANOVAs showed that the pain unpleasantness scores, $F(2, 36) = 12.766$, $P < 0.001$; partial $\eta^2 = 0.415$, but not the intensity ratings, $F(2, 36) = 2.532$; $p = n.s.$; partial $\eta^2 = 0.123$, were statistically significantly different between the valence categories. Adjusted post-hoc tests showed that pain unpleasantness ratings after the negative pictures were significantly higher than after positive ($P < 0.001$), and neutral ($P < 0.002$) pictures. Pain unpleasantness ratings after the neutral pictures were also significantly higher than after the positive primes ($P < 0.021$). Consistent with this, pain unpleasantness ratings showed a significant linear relationship with valence (negative > neutral > positive), $F(1, 18) = 13.611$; $P = 0.002$.

In addition, there was a statistically significant interaction between emotional valence and pain ratings, $F(2, 19) = 5.468$, $P < 0.01$; partial $\eta^2 = 0.365$, indicating that the above-mentioned changes related to picture valence were stronger for pain unpleasantness compared with pain intensity.

We then used descriptive statistics and reported a trend analysis showing a quadratic relationship between pain intensity ratings and picture valence, $F(1, 18) = 4.88$; $P = 0.040$.

Scalp ERP results (N2, P2)

Averaged ERP waveforms and scalp topographies related to each priming condition, time-locked to both the initial visual stimuli and then to the consecutive painful stimuli, are shown in Fig. 2. The painful stimuli in all the three affective conditions elicited ERPs with an early negative peak in the time range between 100 and 150 ms (N2) and a late positive deflection between 220 and 350 ms after the painful onset (P2). The N2 had maximal amplitudes at frontocentral sites peaking at FC2, whereas the P2 had an activity focus at centro-parietal sites, with the

highest amplitudes at CPz. The univariate ANOVA of the mean baseline-to-peak amplitudes of N2 yielded a significant effect of valence, $F(2, 36) = 8.314$, $P = 0.001$; partial $\eta^2 = 0.316$. Adjusted post-hoc comparisons confirmed that the N2 amplitude was higher (more negative) after neutral pictures compared to both, positive ($P = 0.002$) and negative pictures ($P = 0.027$). The ANOVA on the mean baseline-to-peak amplitudes of P2 showed also a significant effect of valence, $F(2, 36) = 5.939$, $P = 0.006$; partial $\eta^2 = 0.248$. Post-hoc comparisons revealed that the P2 amplitude was significantly higher after negative pictures compared to both, positive ($P = 0.005$) and neutral pictures ($P = 0.027$). Consistent with this, P2 also showed a significant linear trend with valence (negative > neutral > positive), $F(1, 18) = 12.498$; $P = 0.002$. A trend analysis for the N2 showed a significant quadratic trend with valence (negative < neutral > positive), $F(1, 18) = 18.193$; $P < 0.001$.

There was no relationship between the changes in the amplitudes of N2 and P2 (Pearson' correlation $r(19) = -0.123$ $P = 0.617$) and no relationship between the changes in the amplitudes of N2 and P2 across picture valence: negative: $r(19) = -0.015$, $P = 0.951$; neutral: $r(19) = -0.143$ $P = 0.561$; positive: $r(19) = -0.134$ $P = 0.585$.

Hierarchical multiple regression analysis

The hierarchical multiple regression was conducted to determine if P2 and then the addition of N2 differential values improve the prediction of pain unpleasantness over and above P2 values alone. The full model including P2 differential values, i.e. $P2_{neg-pos}$ and $P2_{neg-neu}$ (model 1) and N2 differential values, i.e. $N2_{neu-pos}$, $N2_{neu-neg}$ (model 2) to predict perceived pain unpleasantness differences between negative and positive conditions ($UNP_{neg-pos}$) was statistically significant ($R^2 = 0.502$, $F(4, 14) = 3.525$, $P < 0.034$; adjusted $R^2 = 0.359$). The addition of N2 though (i.e. $N2_{neu-pos}$, $N2_{neu-neg}$) to the prediction (model 2) led to a nonstatistically significant increase in R^2 of $= 0.118$, $F(2, 14) = 1.655$, $p = n.s.$

Regarding the prediction of perceived pain unpleasantness differences between negative and neutral conditions ($UNP_{neg-neu}$), the full model including P2, i.e. ($P2_{neg-pos}$ and $P2_{neg-neu}$, model 1) and N2, i.e. ($N2_{neu-neg}$ and $N2_{neu-pos}$, model 2), was statistically significant, $R^2 = 0.481$ $F(4, 14) = 3.242$, $P < 0.044$; adjusted $R^2 = 0.333$. The addition of N2 ($N2_{neu-neg}$ and $N2_{neu-pos}$, model 2) to the prediction led to a nonstatistically significant increase in R^2 of $= 0.119$, $F(2, 14) = 1.610$, $p = n.s.$

We did not attempt to predict perceived pain intensity differences because the intensity ratings did not differ significantly between valence conditions (see Methods section). See Table 1 for full details on each regression model.

Finally, although the normative ratings of arousal differed between the negative and positive pictures we selected for this study, $F(2, 117) = 58.04$, $P < 0.001$, the addition of arousal as covariate in all the above mentioned analyses did not change the significance level of the results (for unpleasantness ratings $P < 0.001$; intensity ratings $p = n.s.$; for P2 $P < 0.001$; for N2 $P < 0.001$).

Brain sources underlying affective modulation of pain

We found multiple neural sources contributing to the emergence of the ERPs (see Fig. 3). In the [400–800 ms] window, which is related to pain processing, we identified eight IC clusters that explained 99.8% of the variance. We approximately located these clusters in the visual cortices, the right somatosensory cortex, the posterior parietal cortex, the angular gyrus, the thalamus, the ACC and the PFC (see Table 2). Interestingly, only cortical sources in the ACC and the thalamus seem to differentiate between picture valence, in the P2 and N2 respectively.

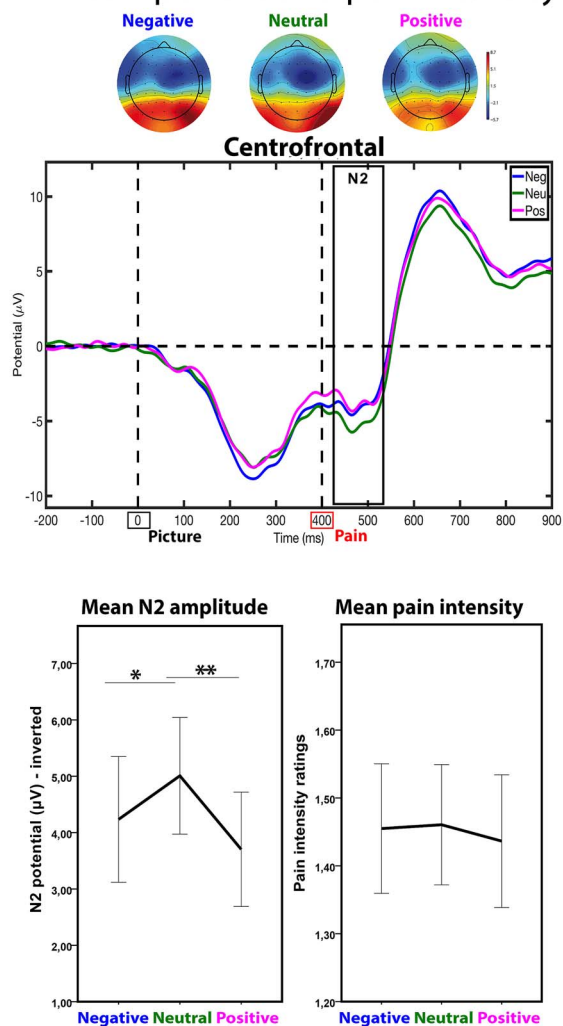
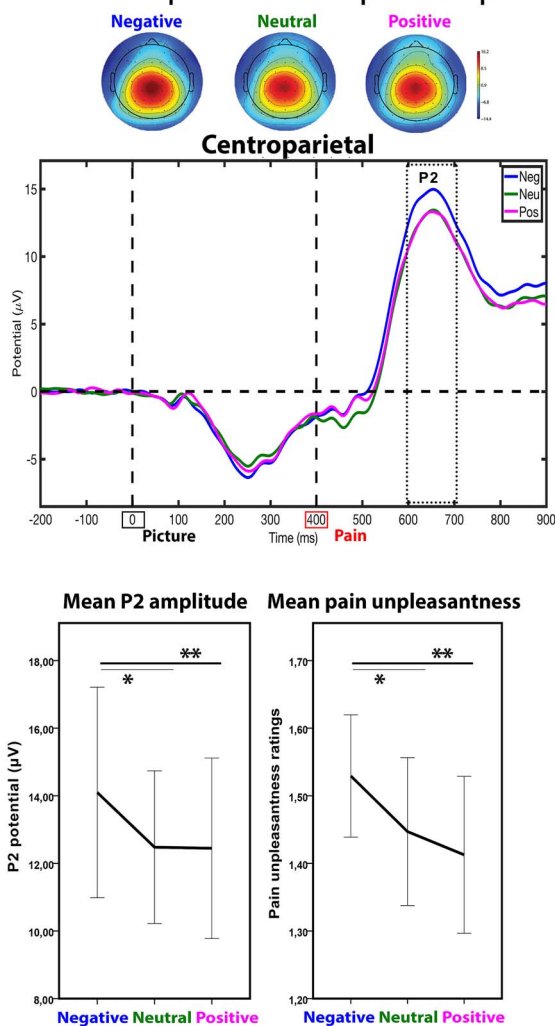
A. N2 component and pain intensity**B. P2 component and pain unpleasantness**

Fig. 2. Scalp ERP and pain ratings. Event-related potentials (ERPs, $n = 19$) evoked by picture onset (0 ms) and pain onset (400 ms). **A.** (top) N2 grand average topographies for latency window (20–120 ms) and N2 ERP waveforms at right centrofrontal cluster sites (Cz, C2, C4, FC2, FC4). (bottom) extracted N2 amplitudes and mean pain intensity ratings across picture category. **B.** (top) P2 grand average topographies for latency window (200–300 ms) and P2 ERP waveforms at centroparietal cluster sites (Cz, C1, C2, CPz, CP1, CP2). (bottom) extracted P2 amplitudes and mean pain unpleasantness ratings across picture category. Error bars reflect SEM. Levels of significance: * $P < 0.05$, ** $P < 0.01$.

Table 1. Hierarchical multiple regression on differential ERP values as predictors of perceived pain unpleasantness. Hierarchical multiple regressions predicting pain unpleasantness differential ratings ($UNP_{neg-pos}$, left and $UNP_{neg-neu}$, right) from differential $P2_{neg-pos}$, $P2_{neg-neu}$, $N2_{neu-pos}$ and $N2_{neu-neg}$ values.

	$UNP_{neg-pos}$				$UNP_{neg-neu}$			
	Model 1		Model 2		Model 1		Model 2	
	B	β	B	β	B	β	B	β
$P2_{neg-pos}$	19.159**	0.841	20.275***	0.889	11.219	0.809**	11.572**	0.834
$P2_{neg-neu}$	-11.576*	-0.603	-18.617*	-0.969	-5.407	3.174	-9.697*	-0.829
$N2_{neu-pos}$			8.571	0.286			3.816	0.209
$N2_{neu-neg}$			1.769	0.299			1.364	0.379
R^2	0.384		0.502		0.361		0.481	
F	4.987*		3.525*		4.529*		3.242*	
ΔR^2	0.384		0.118		0.361		0.119	
ΔF	4.987*		1.655		4.529*		1.610	

* = $P < 0.05$, ** = $P < 0.01$, *** = $P < 0.005$ B, unstandardized regression coefficient. β , standardized regression coefficient.

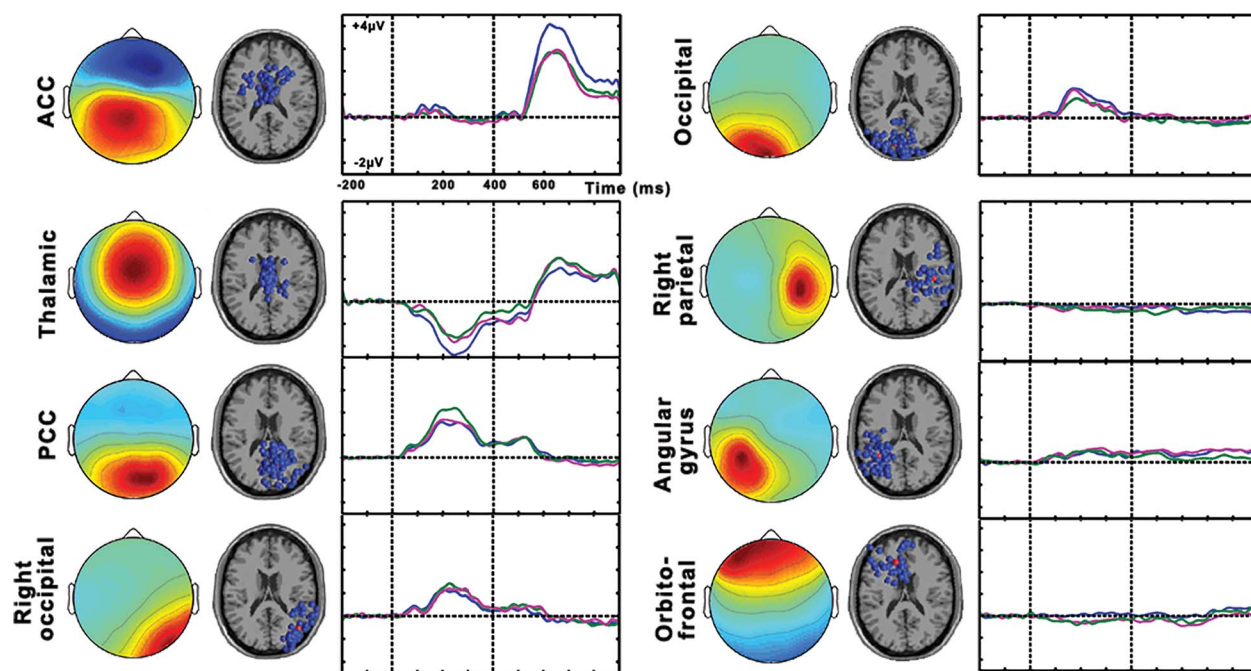


Fig. 3. Independent component (IC) clusters: Mean scalp maps showing distribution of relative projection strengths; dipole source locations (red, centroid; blue, individual dipoles) and related cluster ERPs (blue, negative; green, neutral; fuchsia, positive).

Table 2. Dipole modeling cluster characteristics. Cluster number, cluster name, number of ICs in the cluster, MNI coordinates (x, y, z) of the centroids of each cluster, explained variance (EV) of the cluster respect to the ERP selected window (400–800 ms). Standard deviations are reported in brackets.

Cluster number	Cluster name	Number of ICs	X	Y	Z	EV
1	Thalamus	83	2 (8)	−11 (13)	15 (11)	27.7%
2	Posterior cingulate cortex (PCC)	79	12 (15)	−60 (15)	33 (15)	23.3%
3	Anterior cingulate cortex (ACC)	51	−2 (17)	4 (14)	49 (13)	22.1%
4	Angular gyrus	59	−38 (13)	−50 (16)	35 (21)	11.2%
5	Right occipital	44	44 (13)	−67 (20)	−10 (18)	6.4%
6	Right parietal	46	44 (15)	−24 (19)	54 (14)	4.2%
7	Orbito-frontal	58	−14 (17)	31 (17)	−21 (10)	2.8%
8	Medial occipital	65	−13 (19)	−89 (14)	2 (17)	2.1%

Discussion

Our findings extend an important line of work on the nature of the emotional modulation of pain. We provide neural and perceptual evidence for different mechanisms underlying regulation of pain intensity and unpleasantness. Positive emotional priming reduced the perceived affective dimension of pain, while negative emotional priming amplified it, with the neutral condition falling in between. In contrast, emotional primes had no significant effect on the perceived pain intensity.

In addition, ERPs (N2, P2) to the painful stimulation were also differentially modulated by emotional valence, with higher N2 amplitudes when pain was preceded by neutral compared to negative and positive pictures and higher P2 amplitudes for negative compared to neutral and positive pictures.

The magnitude of the modulation of pain perception could furthermore be predicted only by neural differences in the P2 component between individuals and not by N2.

Pain ratings

We found that pain unpleasantness and pain intensity are differently related to picture valence. Pain unpleasantness ratings were

modulated by picture valence, whereas changes in pain intensity ratings did not reach significance. Emotions seem therefore to modulate pain unpleasantness more than the perceived intensity of pain and is in line with previous findings (Rainville et al. 2005).

Emotional valence modulated perceived pain unpleasantness, with higher unpleasantness ratings after negative pictures compared with neutral or positive pictures. Such differences between the three types of emotional valence (i.e. negative, positive, neutral) were statistically significant. Previous studies reported a similar direction for these modulatory effects, with, however, only a linear trend between emotional category and pain unpleasantness ratings (Kenntner-Mabiala et al. 2008; Kamping et al. 2013). Factors that could explain the differences in the strength of effects between previous studies and the present one may be related to the experimental design. For example, these previous studies (Kenntner-Mabiala et al. 2008; Kamping et al. 2013) applied painful stimuli simultaneously with picture presentation of relatively long duration (6 or 18.3 s). We presented pictures of very short duration (200 ms), which preceded application of painful stimuli.

Time of presentation of pictures relative to electrical stimulus application (preceding or during) might influence pain-specific attentional processes. For example, subject's attention directed

away from the painful stimuli reduced pain ratings (Miltner et al. 1989; Dowman 2001; Kenntner-Mabiala et al. 2008). Similarly, decreased pain ratings were observed when participants' attention to painful stimuli was distracted by picture presentation (Kenntner-Mabiala and Pauli 2005).

Unpleasantness ratings of painful stimuli were higher after negative compared to neutral or positive stimuli, probably due to the similar negative valence (i.e. congruent emotional valence) between negative and painful stimuli, which may have facilitated expectations. These results remained the same after controlling for picture arousal. Cognitive processes such as expectations might have influenced pain perception, although visual stimuli were presented for short duration (200 ms). A previous study showed that the neural processes of expectation occur also for short stimuli (33 ms), albeit slower than for longer stimuli (>500 ms), (Tu et al. 2021). Expectations about a painful event may shift the perceived intensity of pain closer to the expected intensity. The form of expectancy in this context is an automatic process, suggesting a broader, innate mechanism that can modulate pain perception. Although the domain of expectancy covers a broad spectrum, including placebo analgesia, nocebo hyperalgesia, and stimulus expectancy effects, there are fundamental mechanisms that may be influenced by evolutionary processes. These mechanisms shape our anticipation and response to painful stimuli (Atlas and Wager 2012; Peerdeman et al. 2016). Similarly, expectations associated with positive events can decrease pain ratings. For both positive and negative expectations, how top-down expectations interact with bottom-up nociceptive inputs to modulate pain perception remains however a fundamental question (Bushnell et al. 2013; Wiech 2016; Shih et al. 2019).

We found that pain intensity ratings were decreased after positive pictures compared with neutral and negative pictures, however, this difference did not reach statistical significance. Pain intensity ratings also showed a significant quadratic trend with valence (negative < neutral > positive). Using arousal as a covariate, did not change this outcome.

This finding is in contrast to a previous study, which used an experimental design similar to ours, with painful stimuli delivered after picture presentation (Godinho et al. 2006). The authors reported higher pain intensity ratings for unpleasant compared to pleasant stimuli. The discrepancy could be explained by the fact that the authors assessed pain intensity but not pain unpleasantness ratings, which could have collapsed the two dimensions of interest in the same value (Godinho et al. 2006). This view is supported by the apparent inability of untrained participants to distinguish pain into different dimensions (Fernandez and Turk 1994; Chapman et al. 2001).

In the few studies which assessed both intensity and unpleasantness of pain, pain intensity ratings were found to be significantly higher for negative compared with positive material but no agreement regarding neutral, and pain unpleasantness ratings showed a linear decrease from negative to positive stimuli (Kenntner-Mabiala et al. 2008; Kamping et al. 2013). In the present study, we reproduced the results regarding the relationship between pain unpleasantness ratings and picture valence.

Outcomes from these studies highlight the importance of assessing both pain intensity and unpleasantness. Notably, the distinction between the two pain dimensions is not applied in practical settings. For example, in clinical practice, pain is often considered a single construct. This is also true within the context of experiencing pain and its role in social interactions. The distinction between pain intensity and unpleasantness might be

crucial to delineate in clinical pain. Previous studies in chronic pain conditions showed that patients who associate their pain to a greater threat to their health, will rate pain unpleasantness higher than those who associate their pain to a lesser threat to their health, despite having similar self-reported pain intensities (Taenzer et al. 1986; Wade et al. 1990).

EEG results

The mean amplitude of the pain-related ERPs was modulated by affective manipulation. N2 amplitudes were significantly decreased for negative and positive compared to neutral pictures, and P2 amplitudes were larger for negative compared to both positive and neutral pictures.

N2 and P2 amplitudes have previously been shown to be modulated by emotions, however, with inconsistencies regarding the direction and the strength of the modulation. For instance, N2 amplitude was found to be increased for negative compared to positive pictures (Kenntner-Mabiala and Pauli 2005; Kenntner-Mabiala et al. 2008), but also decreased for negative compared to positive or neutral pictures (Ring et al. 2013). In Kenntner-Mabiala et al. (2008), participants were asked to rate pain intensity and unpleasantness, whereas Ring et al. (2013) did not collect any pain-related self-reports. The latter argued that assessment of pain ratings might increase attention-related processes to the painful stimuli. Selective attention to a painful stimulus might therefore differentially modulate pain-evoked brain potentials (Miltner et al. 1989; Fernandez and Turk 1992; Kenntner-Mabiala and Pauli 2005).

In line with this result, a study compared pain-related evoked potentials during application of painful stimuli during viewing of picture of various valence, with a basal condition in which no images were shown (de Tommaso et al. 2009). Compared with the basal condition, N2 amplitude was reduced during the picture viewing, independent of the affective content of the images. The authors interpreted this result as a demonstration of the distracting power of a concurrent visual task while pain was being delivered. This was only true for N2, whereas P2 was similar between neutral and negative pictures, which suggests that N2 and not P2, might be related to attentional effects.

In the current study, N2 amplitude was significantly increased for neutral stimuli compared with both negative and positive stimuli, indicating sensitivity for emotionally arousing stimuli. This finding seems in line with previous studies indicating that changes in N2 are associated with automatic attention allocation to highly arousing stimuli, thereby contributing to memory enhancing of emotion (Hamann 2001; Hajcak et al. 2009).

Modulation of N2 may also be associated with the prepulse inhibition phenomenon (PPI), during which the presentation of pictures of high arousal may have inhibited processing of the pain stimuli. Although not much is known about emotion-induced PPI on pain perception, and associated neural activity, previous findings seem to rather point towards a role of N1 on emotional modulation of PPI. Indeed, the earliest N1 wave may reflect the translation of noxious stimuli into autonomic responses, whereas pain perception could be mediated by later N2 waves (Tiemann et al. 2018).

We found a higher amplitude for P2 for negative compared to positive or neutral pictures, suggesting that P2 could be associated with congruent expectation processes. The display of negative pictures may have generated a negative state, priming the participant to react more negatively to painful stimuli (Wiech 2016). The results on the role of P2 in emotional modulation of pain are quite variable across studies and P2 variation has

been attributed to various processes such as arousal or attention. Previous studies showed that P2 amplitudes were higher during processing of neutral than during processing of negative or positive pictures, although this did not reach statistical significance (Kenntner-Mabiala and Pauli 2005; Kenntner-Mabiala et al. 2008). The authors interpreted their findings by an arousal modulation of P2 due to attention grasping properties of arousing pictures. In line with the attention hypothesis, another study found a decreased P2 amplitude for negative compared to both positive and neutral pictures (Ring et al. 2013).

In line with this hypothesis, differences in arousal from the picture contents may arise from the contents of the affective pictures, which might impact on underlying neural activity and on pain perception (Cook 3rd et al. 1991). Indeed, the startle reflex has been shown to be modulated by arousal, with greatest startle inhibitions occurring for stimuli that are rated as the most arousing (Bradley et al. 1999; Cuthbert et al. 2000; Bradley et al. 2001). It remains however unclear how changes in startle reflex during picture viewing might impact nociceptive modulation and resulting neural activity, and this deserves further investigation.

In the current study, we introduced arousal as a covariate in the analyses to exclude its influence and the results on the amplitudes of N2 and P2 remained the same. Moreover, our findings combined with those of others show, however, that P2 is neither affected by arousal nor by attention-related effects. In addition, another study (Ring et al. 2013) reported differences in P2 amplitude between negative and positive pictures of similar arousal. A previous study also made the attention-related hypothesis unlikely by showing a similar direction of the P2 modulation, regardless of whether participants were paying attention to the intensity or unpleasantness of the pain, or to the pictures presented (Kenntner-Mabiala et al. 2008). Differences in P2 amplitude between previous studies might be related to the various experimental designs, especially the affective priming paradigm used in this study (Richter et al. 2014). Previous studies presented emotional pictures for long durations (6 s) and concomitantly with painful stimuli at various inter stimulus intervals (2.5 or 5 s after picture onset). These designs might induce various higher cognitive processes such as increased attention or working memory that may interfere with pain perception. Affective priming is a rapid and automatic cognitive process, which has been shown to gain magnitude during short time intervals (<500 ms) (Goubert et al. 2003). The choice of a priming paradigm, consisting in experimentally separating pain stimuli and pictures of different valence, can help to avoid explicit cognitive competition between different stimuli and optimize the dissociation between attentional and emotional factors from the pictures and expectation-related factors towards pain. Such an approach might have contributed to the significant relationships we reported between pain ratings and amplitudes of ERPs. Future studies should replicate this experiment using prime pictures of longer durations, and with various time intervals between prime and pain stimuli.

A multiple hierarchical regression, taking into account differences in neural processing as predictors for differences in perceived pain, helped to clarify the relationship between perceived pain and neural activity. Differences in the P2 amplitude ratios between negative and positive pictures and differences in the P2 amplitude ratios between negative and neutral pictures were a good predictor of pain unpleasantness ratings.

Interestingly, differences in the N2 amplitude or its addition to the hierarchical model did not significantly improve the level

of prediction of differences in pain unpleasantness or intensity ratings. Our findings suggest, therefore, that the P2 component may be a neural marker for perceived pain unpleasantness.

Noteworthy, we focused our research on the role of N2 and P2 because they are the most commonly studied pain-related ERPs. Other ERPs might however also be relevant and further work examining their roles in emotional modulation of pain could be of significant importance.

In addition, we applied somatosensory stimuli at a predefined intensity to be in the painful range. The intensity delivered was indeed based on each individual's pain threshold (i.e. 80% between pain threshold and pain tolerance), which should therefore always be rated not only painful but close to the tolerance threshold.

However, the modulation observed on ERPs may have been obtained with other sensory modalities as well, and may therefore not be specific to nociceptive stimulation. To test this hypothesis, we collaborated with Prof. Montoya and Prof. Sitges (University of the Balearic Islands, Spain) to determine the contribution of picture valence on N2 and P2 during nonnociceptive stimulation (tactile). In a previous study, Montoya et al. applied nonnoxious tactile stimuli during presentation of pictures of different valence (pleasant, unpleasant), selected from the IAPS in a sample of 33 healthy female participants (Montoya and Sitges 2006). We contacted the authors and asked permission to examine if the somatosensory N2 and P2 elicited by tactile stimulation could potentially be modulated by picture valence.

We used the same window of interest as the one in the current study, and we showed that somatosensory P2 amplitude was not significantly modulated by picture valence (paired t-test between pleasant and unpleasant valence, $P > 0.46$). Similarly, somatosensory N2 amplitude was not significantly modulated by picture valence (paired t-test between pleasant and unpleasant valence $P > 0.14$). Although this is not a direct control condition to our current study, it shows that the somatosensory N2 and P2 following nonnociceptive stimuli are not modulated by picture valence.

Source analyses showed that the ACC and the thalamus were the main spatial clusters accounting for the difference in the selected time window [400–800 ms], following the application of painful stimuli. These regions have been described as essential for the affective processing of pain (Price 2000).

Cortical sources from the thalamus seem to dissociate negative and positive from neutral valence, and seem to originate in the thalamus, which is in line with the literature showing that thalamic activity is associated with arousal changes (Lane et al. 1997; Liu et al. 2018; Zou et al. 2020).

Changes in ERPs largely explained by ACC sources may finally likely account for augmented pain unpleasantness perception.

In addition, these brain regions are in line with previous studies describing different pathways to process different qualities of pain. For example, tracts connecting the thalamus to the somatosensory cortex are believed to encode the sensory quality of pain (Treede et al. 1999; Rhudy and Meagher 2001; Lyu et al. 2021). Other tracts projecting to the anterior cingulate cortex are believed to be associated with the emotional quality of pain (Price 2000; Hofbauer et al. 2001).

We also found other brain regions (parietal, PFC, angular gyrus), which have been previously described as relevant for somatosensory and pain processing.

The structures with the earliest contribution are occipital and thalamic clusters. Changes in the occipital cortex are consistent with visual activity during picture viewing before the painful

stimulation was delivered, but were not associated with emotional modulation of pain (Foti et al. 2009; Roy et al. 2009).

Further studies investigating the specific elements in the pictures that could predict their impact on pain processing would be a valuable addition to the current study. This could provide a deeper understanding of the mechanisms at play and enhance the practical applications of pain modulation. Future studies are needed to focus on this aspect by employing more detailed analyses of participants' responses to the pictures, such as measuring physiological reactions, emotional responses, and cognitive appraisals. This approach could help identify the key factors that influence how pictures affect pain perception and processing.

Conclusions

We showed that the pain-related temporal neural dynamics of several brain areas was modulated by emotions. Such alterations were also associated with pain ratings. We discussed two mechanisms, namely uncertainty and expectation, which might define the direction of neural and perceptual changes. Moreover, we highlighted individual differences in the emotional modulation of pain, such that the ability to show greater emotional modulation of pain at the perceptual level is related to how emotional discrimination is performed at the neural level.

A better understanding of emotional processes influencing pain perception would help in deepening the knowledge of basic neural and psychological pain-related dynamics. This would also benefit treatment approaches to disorders such as chronic pain, for which a deficient emotional modulation of pain has been described as one of the pathophysiological mechanisms (Klossika et al. 2006; Kamping et al. 2013). Further studies are needed to ascertain the relevance of these interpretations and whether these patterns are differently balanced in clinical population, where maladaptive associations are known to take place.

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Author Contributions

Francesca Zidda (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing—original draft, Writing—review & editing), Yuanyuan Lyu (Data curation, Formal Analysis, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Frauke Nees (Methodology, Supervision, Writing—original draft, Writing—review & editing), Stefan T Radev (Conceptualization, Formal Analysis, Investigation, Methodology, Software, Validation, Writing—original draft, Writing—review & editing), Carol Sitges (Investigation, Methodology, Validation, Visualization, Writing—review & editing), Pedro Montoya (Methodology, Resources, Validation, Visualization, Writing—review & editing), Herta Flor (Conceptualization, Funding acquisition, Investigation, Methodology, Resources, Validation, Writing—original draft, Writing—review & editing), and Jamila Andoh (Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing—original draft, Writing—review & editing).

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